

UZAIR AKBAR

SUMMARY

Software Eng. with experience in Data Eng., Data Science, and DevOps—focused on model deployment, scalable ETL pipelines, and CI/CD. Pursuing an M.Sc. in Computer Science at Georgia Tech, specializing in Robust + Causal ML,[†] with strong interest in Data/ML Eng. and cloud based micro-services. Open to full-time roles starting **Jan. 2026**.

EDUCATION

M.Sc. Computer Science @ Georgia Institute of Technology

Dec. 2025 (expected)

- **Teaching Assistant:** Reinforcement Learning ([CS8803](#), [CS7642](#) Fall '24 — present). **GPA:** 3.81 / 4.0
- **Coursework:** Machine Learning, Systems for ML, Database Systems, Computer Networks, OS, Algorithms

SELECTED PUBLICATIONS

[†]**U. Akbar et al.** “An Analysis of Causal Effect Estimation using Outcome Invariant Data Augmentation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. **(Spotlight; top ~3%)** [🔗](#) [📄](#) [🗣️](#)

PROFESSIONAL EXPERIENCE

Software Engineer Intern @ Hewlett Packard Enterprise, USA

June 2024 — Aug. 2024

- Integrated dynamic KVM machine provisioning ([k8s](#) + [KubeVirt](#)) w/ static ESXi based QA infra. to cut costs.
- Built CI/CD pipeline ([jenkins](#) + [rabbitMQ](#) + [redis](#)) to test, build & release VM templates for QA workloads.

Software Engineer @ OMMAX, Germany

Nov. 2022 — June 2023

- Designed end-to-end Salesforce ETL pipeline ([Airflow](#) + [PostgreSQL](#)), improving data throughput by 58% (by refactoring [python](#) → [postgres](#)) and sales lead identification by 63% (by reducing data redundancy & duplicates).
- Deployed [FastAPI](#) + [Docker](#) based CRM microservices on AWS, including [XGBoost](#)-based ticket classification and RegEx based error code identification.

Data Scientist @ Auto1 Group, Germany

June 2022 — Oct. 2022

- Engineered [LightGBM](#) predictor for car refurbishment costs; €160 MAE decrease (~27.5% improvement) vs. prod.
- Architected data pipelines to streamline model retraining and deployment using S3 + Redshift.

ML Research Intern @ Max Planck Institute of Intelligent Systems, Germany

Apr. 2020 — Feb. 2021

- Proposed a novel use-case of data-augmentation for causal inference – published as my Masters [thesis](#) at TUM.
- Developed a method to reduce hidden confounding bias w/ data-augmentation + instrumental variable regression.

Data Science Intern @ PwC, Germany

Dec. 2019 — Apr. 2020

- Trained Siamese LSTM/CNN ([PyTorch](#)), doc2vec ([Gensim](#)) for internal compliance doc. search/annotation app.
- Deployed the models as micro-services on AWS EKS with Seldon-core (containerization+API endpoint) & [k8s](#).

ML Research Intern @ BMW Group, Germany

July 2019 – Nov. 2019

- Published novel GAN architecture for sim→real augmentation ([ICRA '20](#)), improving segmentation by 6.5% AP.

Data Science Intern @ Allianz, Germany

Oct. 2018 — June 2019

- Built [sk-learn](#) compatible [python](#) package for fraud detection in corporate insurance claims via outlier-detection.
- Implemented features for interpretability + explainability of fraud scores; [lime](#) based feature importance, etc.

TECH. STACK

Data Eng. & Back End: [MySQL](#) [PostgreSQL](#) [Redshift](#) [Redis](#) [Airflow](#) [Spark](#) [FastAPI](#)
DevOps & Cloud: [Git](#) [Bash](#) [Docker](#) [K8s](#) [Jenkins](#) [RabbitMQ](#) [JFrog](#) [AWS](#) (EC2, S3, Redshift, EKS)
ML & Data Sci.: [Python](#) [PyTorch](#) [sk-learn](#) [TensorFlow](#) [LLaMA2](#) [LightGBM](#) [XGBoost](#) [Tableau](#) [C++](#)